

Statistics for machine learning

Mathematical concepts useful for ml. algorithms, things required while deploying

Basic set theory

Combinatory

→ Calculus (knowing derivative & integrals)

imp { × Probability theory
× Statistics

→ needs
Recommended

Programming (MATLAB/Python & R)



Most important → linear algebra (matrices, vectors, inverse / eigenvectors)

app of eigen vectors & values in real-life

Random variable, p.d

joint p.d, covariance, correlation.

probability mass function, bayes theorem,
total probability →
partition of a sample space.

, conditional probability,
conditional independence.

mean, median, mode

Gaussian distri,
Normal distri

why python instead of c, java

→ difficult to deal with high dimensional matrices

even in idla python is X

but it has lots of packages numpy, pandas, scikit, keras?

computer vision tasks

① assign. how to
create matrices

matplotlib
on mat?

deep
learning
tasks

tensorflow,
keras,
pytorch

Images & videos

openCV
Packages

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Data everywhere

Volume of data generated $\uparrow\uparrow$

gen by humans, machine, humans + machine.

Image processing, analyzing & managing this much data.

How to deal with this much data

Data Science

↳ science of analyzing raw data using statistics & ML with the purpose of deriving insights from the data.

Better decisions

inspect, clean, transform, model, analyse, interpret raw data.

DS is multi disciplinary

Who is a data scientist & what are their responsibilities.

Data perspective →
 read data
 data processing & cleaning
 summarizing data
 visualization
 deriving insights from data (machine learning)

Statistics v/s M.L

Problem → data

data → solution

relationship b/w data input & output

See Tomorrow : Learning?

Learning

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28108124

What is intelligence & how intelligence relate to learning

Goal: Solve same problems with the help of machine

Solve $\nearrow$ recognise image using machine w/ ml & dl

Sandam Vallabh Bhai
Pato

& APJ Abdul Kalam.

Image stored in the form of a matrix

$\rightarrow$ pixel count $\times$ pixel count.

to solve

using c, c++, java we've to find how brain does this.

Very complicated,

$\downarrow$
so mimic brain, we learn from xp.

This is where M.L comes into the picture.

& to learn something, we need data.
so data science.

Collect multiple # of images & input them as XYZ image.

M.L. approach $\rightarrow$ instead of prog for every task.

We collect lots of ex that specify the correct output

$\downarrow$
Supervised learning.

an M.L.A then takes these ex & produces a program that does the job.

- ① Prog may look different from typical program.
- ② If done right, it can handle both new cases & training data as well.
- ③ As the data changes so does the m.l.

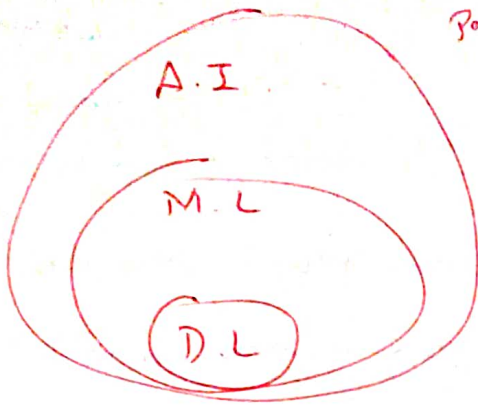
M.L $\rightarrow$ field of study that gives computer the ability to learn
w/o being explicitly programmed.

Definition

Paradigm?

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Learning paradigms

Supervised learning

Unsupervised learning

Semi-supervised learning
reinforcement learning

Applications of M.L.

recommendation

showing ads based on previous preference

biometric systems

stock exchange prediction

Interview Question → I.Q.

Major diff. b/w classification & regression

o/p discrete labels

Problems
classification
prediction
(regression)

o/p: real values

need labelled o/p

output not known

Problems
→ clustering

Prob. distr. estimation

Finding association

Dimension reduction (not all)

not all dim. red. techniques work under unsup. learning

Dimension = feature

if $d \gg \gg \gg 1$ then learning will take some time.

$$\begin{aligned} \mathbf{x} &= [x(1), x(2), \dots, x(d)]^T \in \mathbb{R}^d \\ &\uparrow \\ &\text{sample} \\ &\uparrow \\ &d\text{-dimensional feature vector} \end{aligned}$$

$$\text{Sup. learning: } \{ \mathbf{x}_n \in \mathbb{R}^d, y_n \in \mathbb{R} \}_{n=1}^N$$

Unsup

$$\{ \mathbf{x}_n \in \mathbb{R}^d \}_{n=1}^N$$

no output.

labelled input - output

o/p = can be continuous value or discrete.

Principal component analysis
a linear transformation technique.

H.D. data to lower dim data.

Feature selection is different.

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Which features are important.

We check imp. of each feature w.r.t. to class level.

N.I.A is also important for feature selection.

Genetic algo $\rightarrow$ binary in nature
can be coded into continuous data

2 forms
deal with binary data
deal with continuous data.

~~Random~~ P.S.O., differential algo

Features = $[f_1, f_2, f_3, f_4]$

kind of a subset selection problem.

$[f_1, f_2, f_4]$

Reinforcement learning

data diff from sup & unsup

env. will either reward or penalise.

state, action on env.

then env. will reward or penalise

Semi-sup $\rightarrow$ few labelled samples & many unlabelled samples

data collection is easy
data labelling is tough
(requires expert)

ex, book R.D. Sharma

solved questions $\rightarrow 100$

unsolved questions $\rightarrow 1000$

makes model more generalised /

we should both

consider labelled & unlabelled

data, instead of throwing unlabelled data away

4 elements
possible sub sets $\Rightarrow 2^4 - 1$
as $\emptyset$ is not considered

$2^d - 1$

subsets
choose best one by checking performance.

current buzz field

$\rightarrow 100 \rightarrow$ not sufficient to become expert

diverse

Semi-supervised learning

100 labelled
500 unlabelled

Soln $\rightarrow$ ignore unlabelled

we should ideally not ignore
as training on diverse & large dataset improves
model prediction

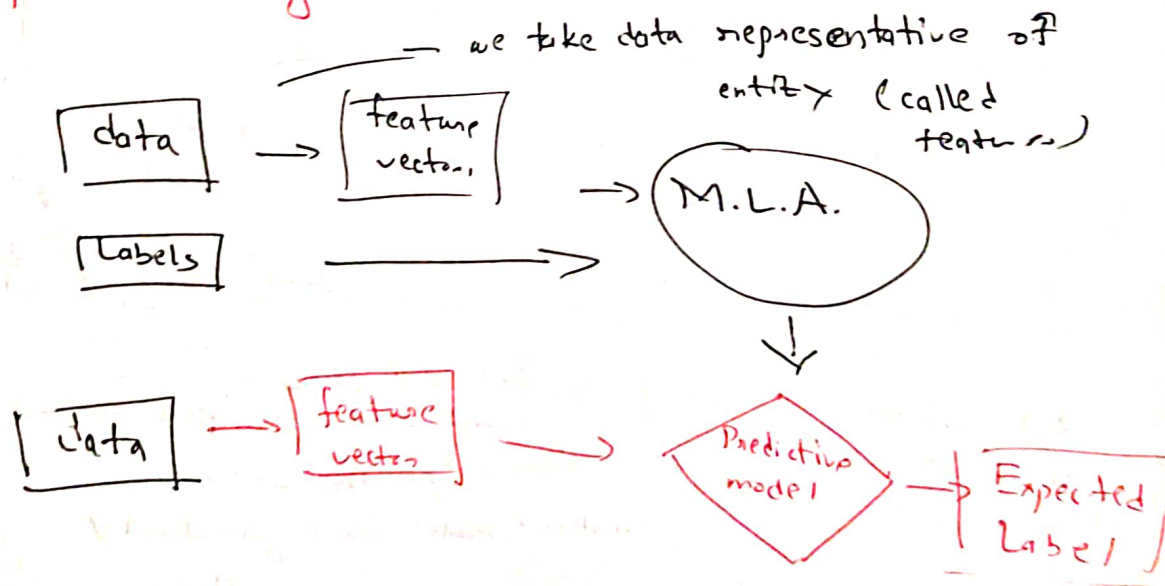
R.D. Sharma solved, unsolved

Continual learning

Train continuously
so that model performs well on new data also.

We only fine-tune the model

Supervised learning structure.



Flower

curman/Readme.md

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sepal_length sepal_width petal_length petal_width species

original data in form of images

converted to this form using attributes

we're to extract the features & then only the learning algorithm work on it

2 stages of developing a model

1. Training

2. Testing

Assignment

iris dataset

(already in scikit)

iris = datasets.load_iris()

matplotlib

Un-Supervised learning

no labels. even in training

data $\rightarrow$ fv $\rightarrow$ M.L.A



data $\rightarrow$ fv $\rightarrow$ Model $\rightarrow$ Likelihood or Cluster id or

There are different types of clustering

Better representation

Learning algorithms examples.

Sup-Learn

Naive Bayes

K-nearest neighbor

S.V.M.

Decision Tree

Random Forest

Feed forward neural network.

Unsupervised learning

k-means

clustering

Principal component analysis

used in deep learning

focus



P.C.A $\rightarrow$ reduce higher dim to low dim.

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Supervised learning example
2 main tasks of sup. learning

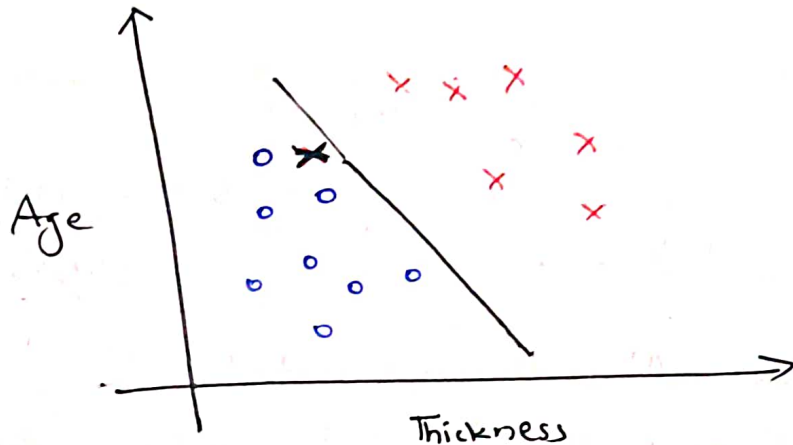
Regression

Classification

2-dimensional feature space

Breast Cancer
(malignant, benign.)

of features may
be more than one.



17 samples (ignore cross on other side)

with 2 features $\rightarrow 17 \times 2$

feature
matrix

age thickness

I want to find out the class label
of the new sample (in black)

check class label of nearest
neighbors

k-nearest neighbors learning
algorithm

① identify # of neighbors we will consider

ex, $k=3$

in 2d
find euclidean distance with all
samples & then sort distance
value in ascending order

first 3 last 3 distance values

& then based on sorting we will get 3 neighbors



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Out of 3 samples, how many belong to class 1 & class 2

here 2 are circle
1 is cross

scikit-learn
dataset

So we assign class circle class.

k not even

↳ evenly split.

tie-cases

To avoid tie-cases we take k-value as odd value

even with k-odd value in case classes > 2

we might have tie-cases

→ Read research article

↓
Assignment

k-nearest neighbor → Lazy learner

↓

steps in
KNN

↓
no training

another is
eager
learning

Learning Paradigm

Lazy
learner

Eager
learner

→ Naive Bayes
↳ Probability of a sample being of a particular class

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

another → separator

Bayes' rule

when → have hypothesis (steve is a librarian)

observed some evidence (neak & tidy soul)

We want $P(\text{Hypothesis given the evidence})$ $P(H|E)$

we're restricting our view only to the possibilities where the evidence holds

$P(H)$ = Probability that hypothesis holds before considering any evidence.

Prion → $P(H) = \frac{1}{1+20}$

Proportion of librarians that fit the description

↑ Likelihood
 $P(E|H) = 0.4$

has much other side of space includes the evidence

→ $P(E|\neg H) = 0.1$
farmers who fit description

$$P(H|E) = \frac{\text{total \# of librarians fitting the description}}{\text{total \# of people fitting the description}}$$

$$= \frac{210 \times P(H) \times P(E|H)}{210 \times P(H) \times P(E|H) + 20 \times P(\neg H) \times P(E|\neg H)}$$

$$= \frac{210 \times \frac{1}{21} \times 0.4}{4 + \frac{210 \times 20}{21} \times 0.1}$$

$$= \frac{4}{4+20} = \frac{1}{6}$$

$$= \frac{1}{6}$$

$$= \frac{P(H) P(E|H)}{P(H) P(E|H) + P(\neg H) P(E|\neg H)}$$

$$= \frac{P(H) P(E|H)}{P(H) P(E|H) + P(\neg H) P(E|\neg H)}$$

$$P(H) P(E|H) + P(\neg H) P(E|\neg H)$$

$$\frac{P(H) P(E|H)}{P(E)}$$

$P(E)$ $\rightarrow$ total probability of seeing the evidence.

$$P(H|E)$$

$\uparrow$
Posterior

belief about the
hypothesis after seeing
the evidence

Classifications --

- ① KNN,
- ② tree based (decision tree & random forest)
- ③ Probability,
- ④ discriminant function based

(neural networks, &

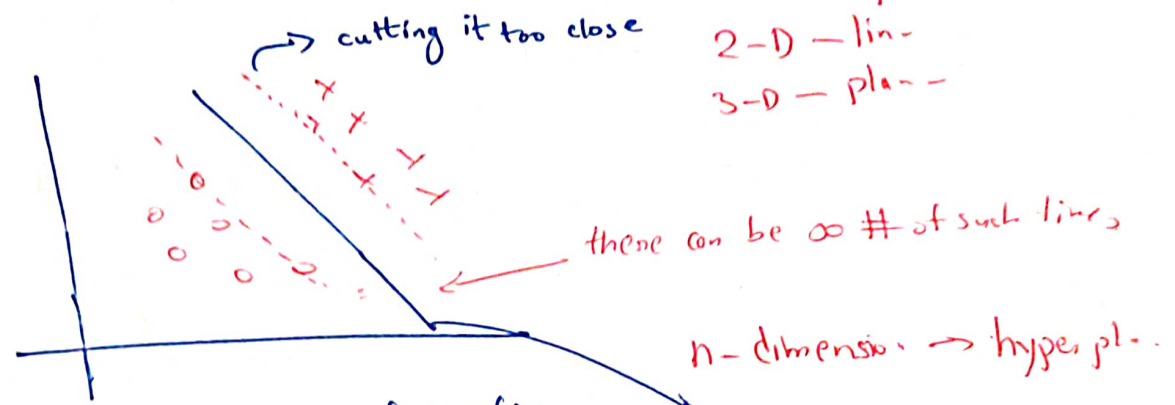
Support vector machine)

They usually get a decision boundary.

Separation no. ---

2-D - line -

3-D - plane -



other types of classific,

ensemble,

rule based.

What is a decision boundary, how to draw it.

although ∞ D.B. can be drawn, we've to choose best out of them.

S.V.M gives us optimum D.B.

$$y = mx + c$$

$\uparrow$ slope $\uparrow$ y-intercept

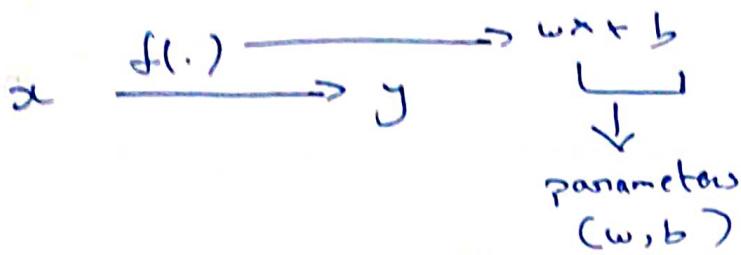
$$y = wx + b$$

$\uparrow$ weight $\uparrow$ bias

slope decide the orientation of hyperplane.

$b \rightarrow$ position of hyperplane.

if $b = 0$, hyperplane will pass through origin.



The f relies on 2 parameters w & b .

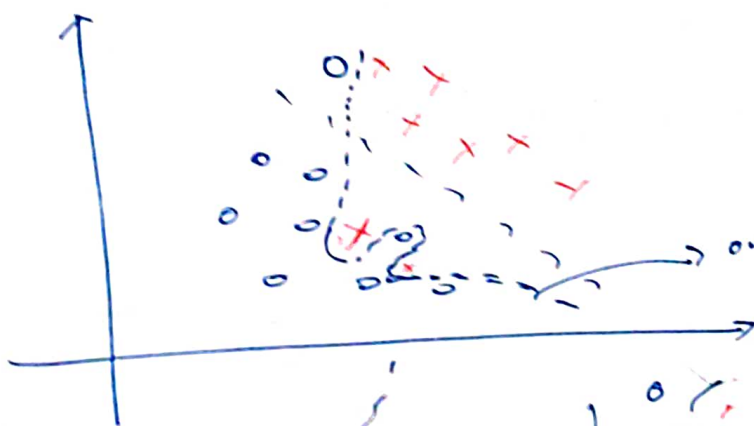
Training = searching for relation
finding values of w & b

Linearly separable $\rightarrow$ definition

w can be a vector
 w can be a scalar

size of w = size of x

Point lying on one side c_2
below c_1



can't be separated linearly.

overfitting / over training
fitting model to noisy samples

Training performance
is very good
but test performance
is very bad

called
overfitting



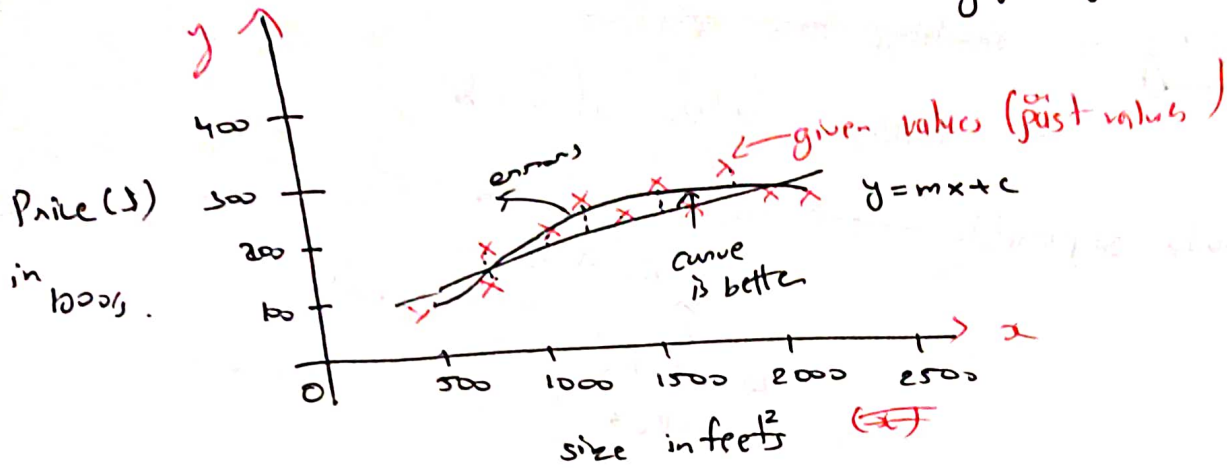
These samples are called
noisy samples.

Ignore noisy samples.

error $\rightarrow 0$ not $\Rightarrow$ error $= 0$

Supervised learning example

Housing price prediction

Supervised learning
"right answer," given.

Regression: predict continuous valued output (price)

We want to find house price for any given size.

error $= \sum$ distance of a point not on line to line

M.S.E (Mean Square Error)

linear regression $\rightarrow$ linenon-linear regression $\rightarrow$ curve

x	y
150	—
520	—
600	—
480	0

given

Assignment

can kNN be applied for regression task.

Yes, How?

look at neighbors & apply average rule

Task:

To summarise whole course in ^{any} way $\rightarrow$ can ^{that} classification algo be used for regression?
 (Instead of majority rule)

Unsupervised learning examples:-

market segmentation, friends grouping

Database mining $\rightarrow$ large datasets from growth of information
ex, web click data, medical records, billing, $\rightarrow$ handwriting recog. systems
Applications that we can't program by hand
self-customizing programs.

$\hookrightarrow$ recommendations

USPS can sort letters based on handwriting.

What computers see?

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A digital image

we need pre-processing before we can feed a data into an algo.

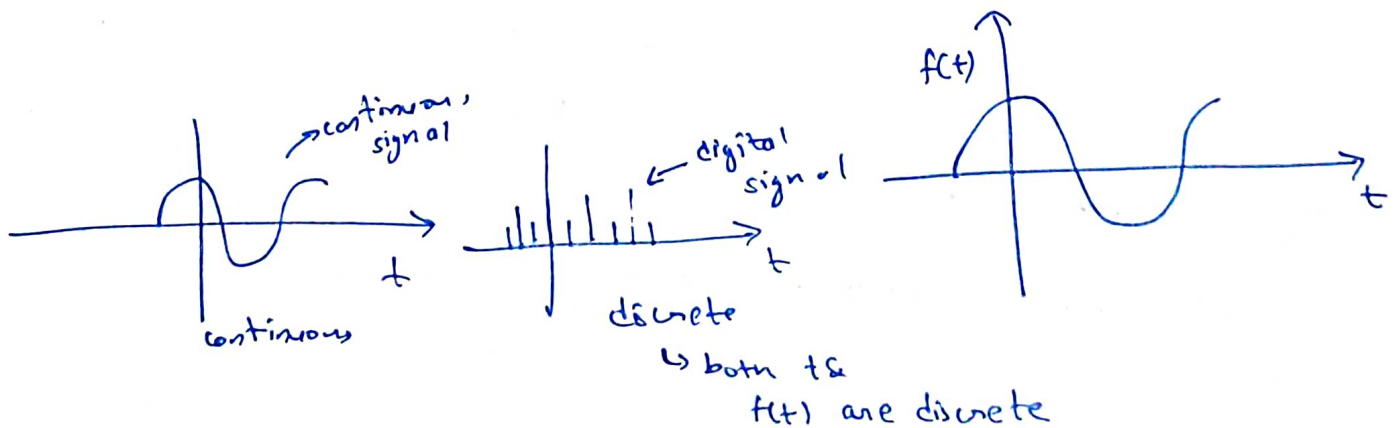
image processing

image = 2-dimensional ftn

signal = 1-D ftn

signal always varies with time

amplitude varies with time.



digital images $\rightarrow f(x, y)$ x, y coordinates.

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Who is converting analog image into digital image.

sampling & quantization

$$\frac{2^8 \times 2^8 \times 2^8}{\text{---}}$$

$\rightarrow 2^{24} \rightarrow 24 \text{ bit image}$

R, G, B

$\downarrow \quad \downarrow \quad \downarrow$

$n \times m \quad n \times m \quad n \times m \rightarrow n \times m \times 3$

$\rightarrow 3\text{-value} \quad 2^{24} = 16 \text{ million}$

$$f(x, y) = \begin{bmatrix} \text{value } f(0,0)_R, f(0,0)_G, f(0,0)_B \\ f(0,0) \quad f(0,1) \quad \dots \quad f(0,M-1) \\ \vdots \\ f(n-1,0) \quad \dots \quad f(n-1,M-1) \end{bmatrix}$$

image definition $\rightarrow$ as a 2D matrix where each item represent R, G, B val.

For 2-co-ordinates $f(x, y)$ v/s (x, y)

Colour of pixel of any image depends on

$$f(x, y) = r(x, y) \times i(x, y)$$

$\downarrow$
reflectance $r(x, y)$
of the object
on which
light is
falling

illumination
 $i(x, y)$ of the
light source

image formation process

light source & reflectance property of the object.

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Image must be represented in form of feature vector

we must first find a form which is representative of the image.

if every pixel ~~has~~ is represented

$$1024 \times 1024$$

$$\rightarrow 2^{10} \times 2^{10} = 2^{20}$$

$$\frac{1024}{2^{10}} \times \frac{1024}{2^{10}} = 1$$

in an image neighborhood relationships are imp.

2D $\rightarrow$ 1D this info is going to be lost

$\therefore$ we're not interested in using pixels as features.

& are interested in finding representative features.

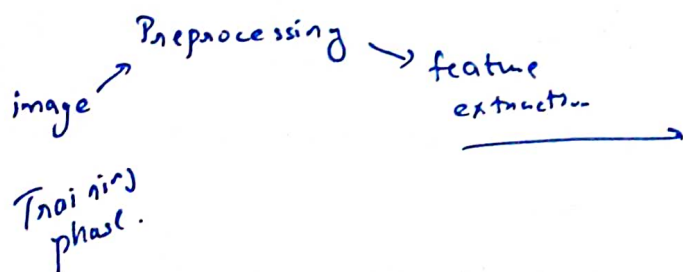
$\therefore$ extract features representative of image.

Feature set

$$x_1 = [f_1, f_2, f_3, \dots, f_d]^T \in \mathbb{R}^d$$

$$x_2 \dots$$

can be
SVM,
LDA,
neural
networks
etc.



Training examples

Feature vector 1

tiger

$\vdots$

fv 2

tiger

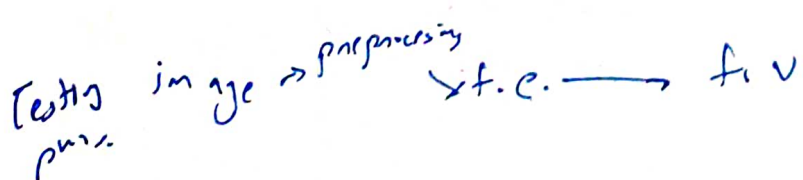
$\vdots$

fv 3

be

can put 1st stage
called
feature
selection
select good
features &
remove bad

$\rightarrow$ classification $\rightarrow$ class label



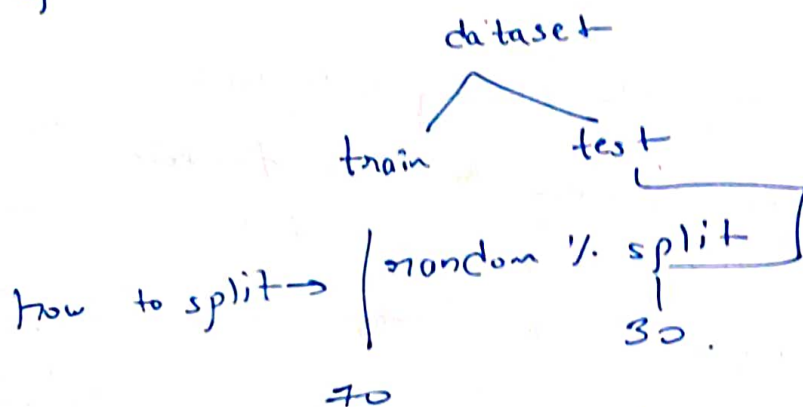
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Good features $\rightarrow$ easily distinguish

bad feature $\rightarrow$ not distinct & irrelevant

M.L $\rightarrow$ 2 phases $\rightarrow$ training $\uparrow$ testing
 build model

① split data set into train set, test set.



Evaluation measure

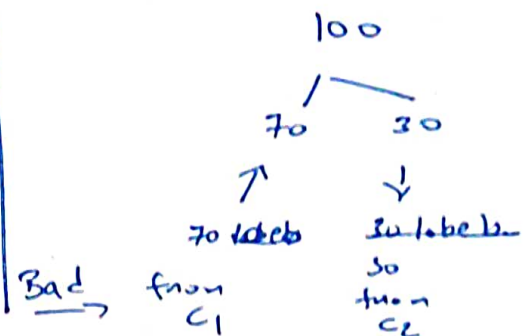
success: actual output $\rightarrow$ target

error: actual o/p $\neq$ target

$$\% \text{ error} = \frac{\# \text{ errors}}{\# \text{ samples}}$$

class 1 = 60 samples

class 2 : 40 samples



both C_1 & C_2 should be equal
 70 from every class

This is called stratified random % split

C_1 60 $\rightarrow$ 70% $\rightarrow$ 42

C_2 40 $\rightarrow$ 70% $\rightarrow$ 28

randomly choose 42 from 60
 28 from 40 } random

Randomness $\therefore \rightarrow$ first 42

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18 $\rightarrow$ might be have some imp. feature.

To avoid biasness in data we take data stratified

$\hookrightarrow$ fairness.

model will get biased towards one class

$\hookrightarrow$ called **biase problem** 11.1

Load this dataset, print feature, ⁽¹⁾ ⁽²⁾ split dataset into train set & test set ⁽³⁾

Evaluation measures

Goals while designing an ml model

$\hookrightarrow$ ① good in training

② good in testing

Training: model is built

error on training data
- training error

Testing: model is applied to the new data similar testing error.

Generalization $\rightarrow$ ^{how well} model performs on new data

performs well on new data $\rightarrow$ good generalization

Test error $\rightarrow$ generalization error

Q how is generalization different related to overfitting?



Training perf $\uparrow$

Testing perf $\downarrow$

model gets fit to noise also

Next: over/under fitting, how to avoid, validation set.

Overfitting (Overtraining)

- ① Low training error + High test/generalization error.
- ② Fit to the noise in the training data.

Training
95%.

Test
50%.

Overfitting = Poor generalization

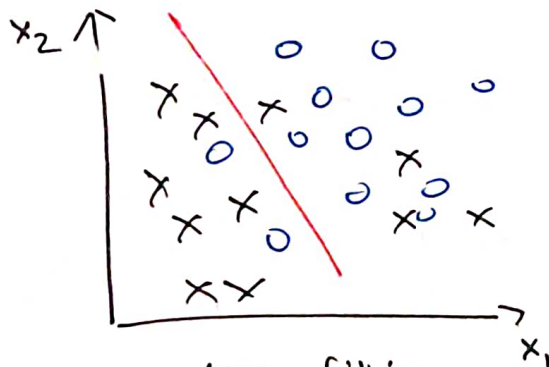
Underfitting (Undertraining)

High ~~train~~ training error & high test/generalization error.

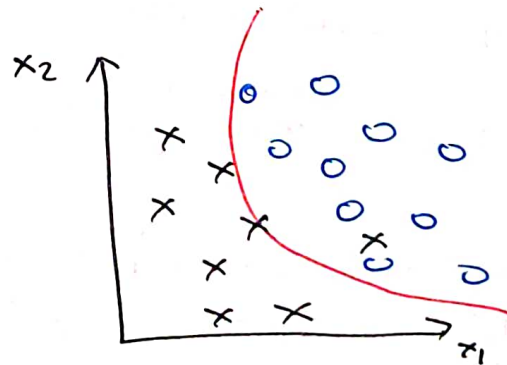
Can't expect model to perform good on test data

Soltn: train model till convergence.

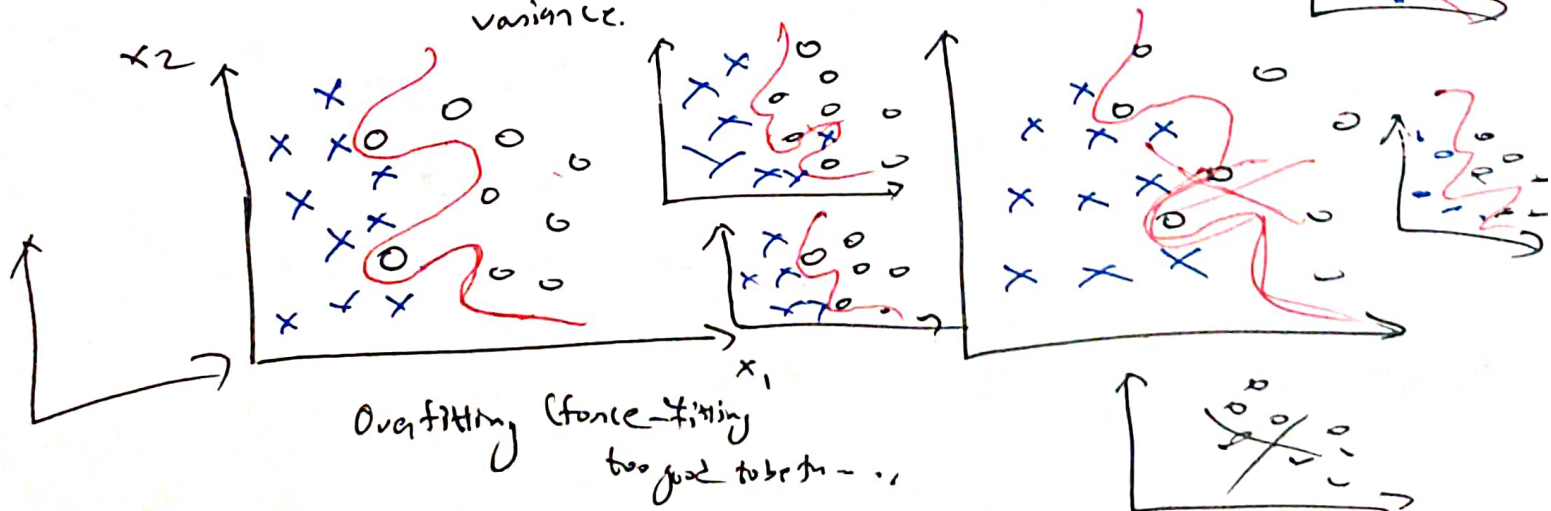
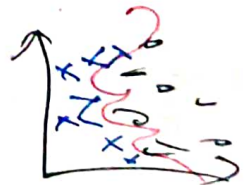
Cause of poor performance: over/under-fitting.



Under-fitting
too simple explain
variance.



Appropriate
fitting



Overfitting (curve-fitting)
too good to be true...

Cause of overfitting

model is too complex, too many parameters.

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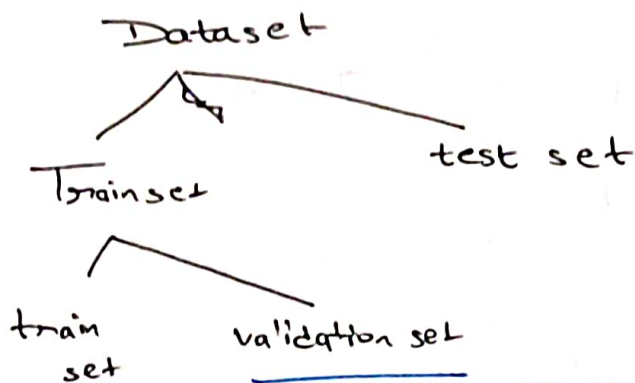
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How to prevent overfitting?

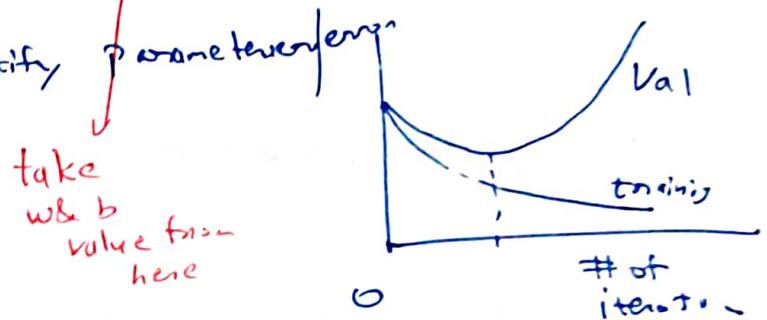
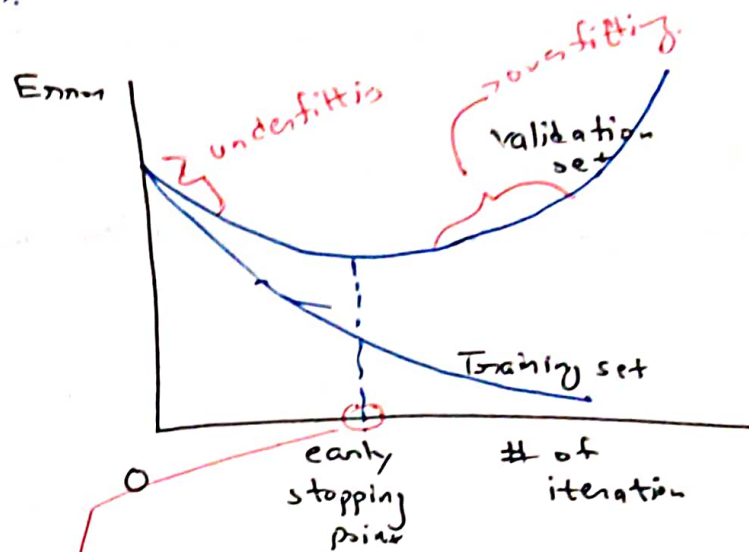
use regularization techniques.

① Early stopping.

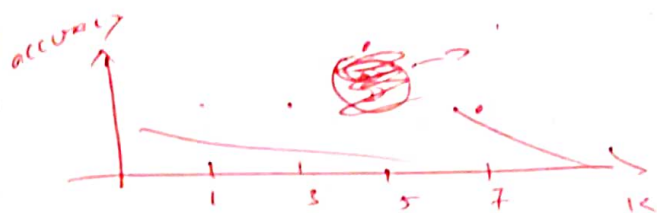
validation set



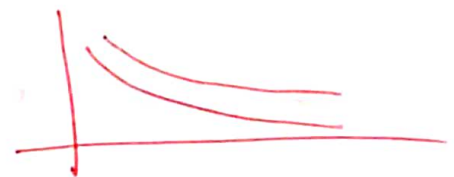
used to identify



Can use this to find k for KNN



ideal



Thus, validation set can be used to find parameters of a classifier.

Confusion matrix

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There are multiple perf. matrix / measure,
metrics,
available specifically for classification.

For classification Confusion Matrix is used

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Confusion matrix $\rightarrow$ a tool to determine the performance of
classifier. It contains info about actual & predicted classifications.

		Predicted	
		+ve	-ve
Actual	+ve	TP	FN
	-ve	FP	TN

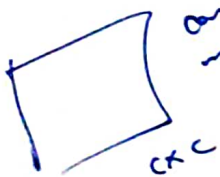
Accuracy of a classifier / model

Binary classification $\rightarrow$ 2 classes

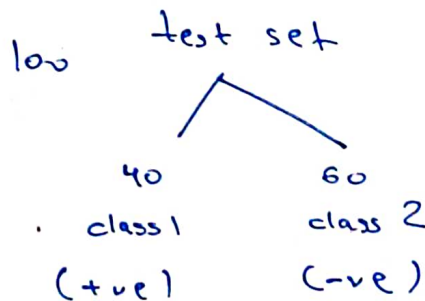
+ve -ve

When we say perf. of model, we mean perf. on test set.

C $\rightarrow$
classes



confusion
matrix



		Predicted		
		+ve	-ve	
Actual	+ve	35 TP	5 FN	40
	-ve	10 FP	50 TN	60

Correctly classified = 85

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$$\begin{aligned} \text{TPR} &= \frac{TP}{TP+FN} & 1 - \text{FNR} \\ \text{True positive rate} & & \\ \text{(sensitivity)} & & \\ \text{(recall)} & & \end{aligned}$$

$$\begin{aligned} \text{True negative rate} &= \frac{TN}{TN+FP} & 1 - \text{FPR} \\ \text{(specificity)} & & \end{aligned}$$

$$\text{accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

$$\text{precision} = \frac{TP}{TP+FP}$$

$$F_1 \text{ score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= \frac{2 \times \left(\frac{TP}{TP+FP} \right) \times \left(\frac{TP}{TP+FN} \right)}{\frac{TP}{TP+FN} + \frac{TP}{TP+FP}}$$

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Ideal model better specificity & sensitivity

Accuracy
Sensitivity
Specificity

$$\left. \begin{array}{l} \text{Accuracy} \\ \text{Sensitivity} \\ \text{Specificity} \end{array} \right\} 0 < \sim < 1$$

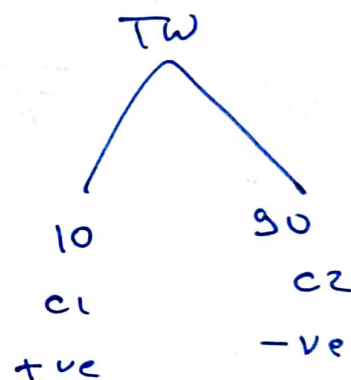
Sensitivity more imp for medical data

		predicted.	
		+ve	-ve
actual	+ve	TP 0	10 FP
	-ve	0 FP	90 TN

$$\text{accuracy} = \frac{90}{100}$$

$$\text{specificity} = 0$$

$$\text{sensitivity} = \frac{90}{90} = 1$$



Balanced accuracy

$$= \frac{\text{sens} + \text{spec}}{2}$$

$$= \frac{0 + 1}{2} = 0.5$$

TASK: Confusion matrix for > 2 classes

Assignment
Design CM
for $C > 2$

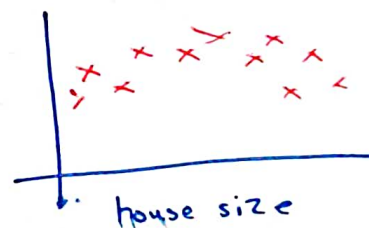
	C_1	C_2	C_3
C_1			
C_2			
C_3			

if focusing on C_1 , $C_1 \rightarrow$ T.P.
others $\rightarrow$ ~~not~~ not C_1

F.N = type II error

F.P $\rightarrow$ type I error

house price



Regression Error Metrics

Mean Absolute Error =

$$M.A.E = \frac{1}{n} \sum |y - \bar{y}|$$

mean square error.

$$M.S.E = \frac{1}{n} \sum (y - \bar{y})^2$$

$\uparrow$ easier to compute than M.A.E
 $\uparrow$ actual $\downarrow$ predicted

why square
the $+/-$ ve
will
cancel out

we don't want
this we want
to reduce error.

$$\text{Root MSE} = \sqrt{M.S.E}$$

Probability

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Random experiment



coin toss, dice throw

total possible outcomes known

but occurrence of an outcome is unknown

[Total possible outcomes are known but the occurrence of a particular outcome is uncertain]

Ex, tossing a coin, rolling a die

Probability = Measure of uncertainty of such a random experiment

$$P = \frac{\text{no. of favorable outcomes}}{\text{total no. of outcomes}}$$

Sample space (S) = contains all possible outcomes of a random experiment

Event = (E) $E \subseteq S$

Types of events: Sure event $P(E) = 1$

Impossible event $P(E) = 0$

Mutually exclusive event $E \cap F = \phi$ (phi)

Exhaustive event $E \cup F = S$

ex) M. Ex. Event

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Exhaustive event

$$S = \{1, 2, 3, \dots, 100\}$$

E_1 = event of getting even no

$$= \{2, 4, \dots, 98, 100\}$$

$$= 50$$

E_2 = odd

$$= \{1, \dots, 99\}$$

$$= 50$$

$$E_1 \cup E_2 = S \quad | \quad E_1 \cap E_2 = \phi$$

HH, TH,
HT, TT.

Rules

Events

A & B

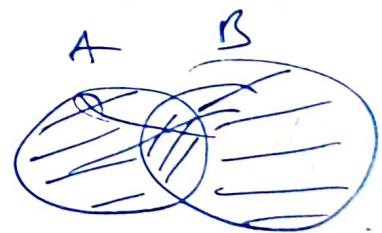
→ set theory

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$n(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$n(A) + n(B) - n(A \cap B)$$

n = cardinality = P



$$\frac{n(A \cap B)}{n(S)} = \text{Probability of } A \cap B$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) \\ - P(A \cap B) - P(B \cap C) - P(A \cap C) \\ + P(A \cap B \cap C)$$

★ Conditional probability

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$E, F \subseteq S$

$A \subseteq B$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad P(B) \neq 0$$

$P(A)$ ~~not~~ given

$P(B)$ has occurred

$$P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{P(A)} \quad P(A) \neq 0$$

Next: BAYE'S THEOREM

$$P(A|A) = \frac{P(A \cap A)}{P(A)} = \frac{P(A)}{P(A)} = 1$$

$$P(S|A) = 1$$

Multiplication theorem

$$P(A \cap B) = P(A) \cdot P(B|A) \\ = P(B) \cdot P(A|B)$$

Independent events

Occurrence of one doesn't affect the occurrence of other

$$P(E \cap F) = P(E) \cdot P(F)$$

$$P(E|F) = \frac{P(E \cap F)}{P(F)} = \frac{P(E) \cdot P(F)}{P(F)} = P(E)$$

Partition of sample space:

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☆ Set of events,

$$\{E_1, E_2, \dots, E_n\}$$

is ~~set~~ said to be a partition of sample space

if i) E_i 's are exhaustive events

ii) E_i 's are mutually exclusive

$$E_1 \cup E_2 \cup \dots \cup E_n = S$$

$$E_i \cap E_j = \emptyset \\ \forall i \neq j$$

Theorem of total probability $\rightarrow$ NEXT

read what is total probability

Total probability

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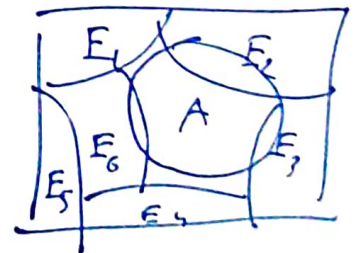
$$\{E_1, E_2, \dots, E_n\} \quad S$$

└──────────┘
partitions

$$A \subset S$$

$$P(A) = P(E_1)P(A|E_1) + P(E_2)P(A|E_2) + \dots + P(E_n)P(A|E_n)$$

$$= \sum_{i=1}^n P(E_i)P(A|E_i)$$



$$P(A) = P(A \cap S)$$

$$= P(A \cap (E_1 \cup E_2 \dots E_n))$$

$$= P(A \cap E_1 \cup A \cap E_2 \cup \dots \cup A \cap E_n) = \sum P(A \cap E_i)$$

$$P(A \cap B) = P(A) \times P(B/A)$$

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Bayes theorem

$$P(E_i/A) = \frac{P(E_i) P(A/E_i)}{\sum_{j=1}^n P(E_j) P(A/E_j)}$$



$$\frac{P(E_i \cap A)}{P(A)} = \frac{P(E_i) P(A/E_i)}{P(A)}$$

Binary classification task

class c_1 c_2
 $\uparrow$ $\uparrow$
 E_1 E_2

$$P(c_1/x) = \frac{P(c_1) P(x/c_1)}{P(A)} \rightarrow \text{ignored}$$

$$P(c_2/x) = \frac{P(c_2) P(x/c_2)}{P(A)}$$

↑
test
sample

given a test sample x , possibility of x being c_1

This is called Posterior probability
 $P(c_1/x)$

$P(c_1)$ Prior probability
 $P(c_2)$ given already

$P(c_1) P(x/c_1)$ class conditional probability
 Probability of x given c_1

$P(x)$ = Probability distribution

if $P(c_1/x) > P(c_2/x)$

$\hookrightarrow c_1$

else

c_2

Random variable

$S = \{HH, HT, TH, TT\}$

R.V. fn that assigns numerical value to each outcome

X : # of heads present

$\uparrow$

R.V.

always cap. for

$X: 2, 1, 0, 1$ — discrete R.V.

$P(X): \frac{1}{4}, \frac{1}{2}, \frac{1}{4}$

Probability distribution

$\sum P(X) = 1$

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wt value of 1000 students

D.R.V $\rightarrow$ Probability Mass FunctionC.R.V $\rightarrow$ Probability Density Function

$$P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B)}$$

$$P(0) \xrightarrow{\text{notation}} P(x_1) = P(x=x_1)$$

$$P(1) = P(x=x_1)$$

$$P(x_i) = P(x=x_i)$$

How to find if P.D is P.M.F or P.D.F

$$\text{PMF} \rightarrow 0 \leq p(x_i) \leq 1 \quad \sum p(x_i) = 1$$

Cumulative Distribution Function (C.D.F)

$$p(X \leq x) = \sum_{x_i \leq x} p(x_i)$$

$$P(X \leq 1) = P(0) + P(1)$$

Continuous R.V.

The P.D. for C.K.V. is
P.D.F

 $f(x)$

Probability distribution.

$$\rightarrow P(x) = \frac{1}{10}x$$

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PCN

→ P.M.F

НН, ТН, НТ, ТТ

$$f(x) \geq 0 \quad \forall x \in R_x$$

$$\int_{\mathbb{R}^n} f(x) dx = 1 \quad \int_{-\infty}^{\infty} f(x) dx = 1$$

$$f(x) = 0 \quad \text{if } x \notin \mathbb{R}_+$$

Cumulative D.F. $\rightarrow$ Distribution

$$p(X \leq x) = \sum_{x_i \leq x} p(x_i)$$

$$p(X \leq x) = \int_{-\infty}^x f(x) dx$$

1. 0.3
2. 0.3
3. 0.3
4. 0.3
5. 0.3
6. 0.3
7. 0.3
8. 0.3
9. 0.3
10. 0.3
11. 0.3
12. 0.3
13. 0.3
14. 0.3
15. 0.3
16. 0.3
17. 0.3
18. 0.3
19. 0.3
20. 0.3
21. 0.3
22. 0.3
23. 0.3
24. 0.3
25. 0.3
26. 0.3
27. 0.3
28. 0.3
29. 0.3
30. 0.3
31. 0.3
32. 0.3
33. 0.3
34. 0.3
35. 0.3
36. 0.3
37. 0.3
38. 0.3
39. 0.3
40. 0.3
41. 0.3
42. 0.3
43. 0.3
44. 0.3
45. 0.3
46. 0.3
47. 0.3
48. 0.3
49. 0.3
50. 0.3
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67. 0.3
68. 0.3
69. 0.3
70. 0.3
71. 0.3
72. 0.3
73. 0.3
74. 0.3
75. 0.3
76. 0.3
77. 0.3
78. 0.3
79. 0.3
80. 0.3
81. 0.3
82. 0.3
83. 0.3
84. 0.3
85. 0.3
86. 0.3
87. 0.3
88. 0.3
89. 0.3
90. 0.3
91. 0.3
92. 0.3
93. 0.3
94. 0.3
95. 0.3
96. 0.3
97. 0.3
98. 0.3
99. 0.3
100. 0.3

$$0 \times \frac{1}{4} + 1 \times \frac{1}{2} + 2 \times \frac{1}{4}$$

— — 1

~~★~~ Expectation of R.V.
(mean / average)

$$E[X] = \sum_x x p(x) \text{ for D.R.V.}$$

↓

mean of X

$$\int_{-\infty}^{\infty} x f(x) dx \text{ for C.R.V.}$$

$$E[g(x)] = \sum_x g(x) p(x) \quad \text{D.R.V.} \\ g(x) = X$$

$$\int_{-\infty}^{\infty} g(x) f(x) dx \quad \text{C.R.V.}$$

$$\therefore E[X^2] = \sum_x x^2 p(x)$$

Properties of expectation.

$$E[ax+b] = E[ax] + E[b] \\ = aE[X] + b$$

$$\Rightarrow \sum_x (g(x)) p(x)$$

$$\sum_x (ax+b) p(x)$$

$$\sum_x axp(x) + \sum_x b(p(x))$$

$$= a \sum_x xp(x) + b \sum_x (p(x))$$

$$= a \sum_x xp(x) + b(1)$$

$$= a \sum_x xp(x) + b$$

$$= aE[X] + b$$

$$E[X_1 + X_2 + \dots + X_n] = E[X_1] + E[X_2] + \dots + E[X_n] \\ \text{(Linearity property)}$$

$$f(x) = \frac{x+2}{25}$$

$$x = 1, 2, 3, 4, 5$$

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$$E(g(x)) = \sum_x g(x) p(x)$$

$$1 \frac{3}{25} + \frac{2 \cdot 4}{25} + \frac{3 \cdot 5}{25} + \frac{4 \cdot 6}{25} + \frac{5 \cdot 7}{25}$$

$$= \sum_x \frac{x+2}{25} p(x)$$

Correct

$$\frac{1}{25} (25)$$

$$= \left(\frac{1+2}{25}\right) \frac{1}{5} + \left(\frac{2+2}{25}\right) \frac{1}{5} + \left(\frac{3+2}{25}\right) \frac{1}{5} + \left(\frac{4+2}{25}\right) \frac{1}{5} + \left(\frac{5+2}{25}\right) \frac{1}{5}$$

$$\frac{3}{25} + \frac{8}{25} + \frac{15}{25} + \frac{24}{25} + \frac{30}{25}$$

$$= \frac{1}{5} \left(\frac{3}{25} + \frac{8}{25} + \frac{15}{25} + \frac{24}{25} + \frac{30}{25} \right)$$

$$\frac{1}{25} (11 + 15 + 59)$$

$$= \frac{1}{5} \left(\frac{7+11+7}{25} \right)$$

$$\frac{1}{25} (70 + 15)$$

$$\frac{85}{25}$$

$$= \frac{1}{5} \left(\frac{25}{25} \right)$$

$$= 1/5$$

$$f(x) = \begin{cases} x & 0 < x < 1 \\ 2-x & 1 \leq x < 2 \\ 0 & \text{otherwise} \end{cases}$$

$$E(x) = \int_{-\infty}^{\infty} x f(x) dx$$

$$= \int_0^1 x^2 dx + \int_1^2 x(2-x) dx = \left[\frac{x^3}{3} \right]_0^1 + \left[x^2 - \frac{x^3}{3} \right]_1^2$$

$$= \frac{1}{3} + 4 - \frac{8}{3} - 1 + \frac{1}{3}$$

$$= 3 - \frac{8}{3} = 3 - 2 = 1$$

Variance of R.V.

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$$\begin{aligned} V[X] &= E[(X - \mu)^2] \\ &= E[(X - \mu)^2] \\ &= E[X^2] - (E[X])^2 \end{aligned}$$

$$E(X^2 - 2\mu X + \mu^2)$$

$$E(X^2) - 2\mu E(X) + E(\mu^2)$$

$$E(X^2) - 2\mu \cdot \mu + \mu^2 E(1)$$

$$E(X^2) - \mu^2$$

$$V(ax+b) = a^2 V(x)$$

$$V(\alpha x) = \alpha^2 V(x)$$

$$V(x-a) = 1 \cdot V(x) = V(x)$$

$$f(x) = \frac{x}{15} \quad x = 1, 2, 3, 4, 5$$

$$E(x) \quad V(x)$$

$$E(x) = \sum_x g(x) p(x)$$

$$= \sum_x \frac{x}{15} \cdot p(x)$$

① Discrete Probability Distribution

- Binomial dist.
- Poisson

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② Continuous Probability Distribution

- uniform
- exponential
- Gaussian ~~dist.~~ (Normal Probability Distribution)

Poisson

$$f(x) = P(X=x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

$x = 0, 1, \dots, \infty$
 $\lambda > 0$

Poisson distribution is used for modelling experiment having waiting as an occurrence.

$$E[X] = \lambda$$

$$V[X] = \lambda$$

Uniform

$$f(x) = \frac{1}{b-a} \quad a < x < b$$

0 otherwise

for $a < x < b$
we'll get uniform values.

$$E[X] = \frac{a+b}{2}$$

$$V[X] = \frac{1}{12}(b-a)^2$$

Exponential

$$f(x) = \begin{cases} \frac{1}{\lambda} e^{-x/\lambda} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

$$E[X] = \lambda$$

$$V[X] = \lambda^2$$

~~Gaussian / N~~

Gaussian / Normal

~~Gaussi~~
~~Gau~~
~~Q~~

$$f(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2}$$

$$\sigma > 0$$

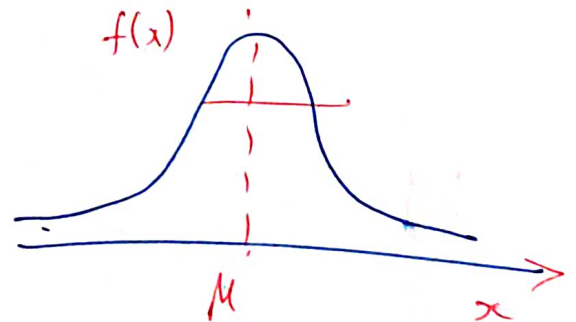
~~$$-\infty < x < \infty$$~~

$$-\infty < x < \infty$$

$$E[X] = \mu$$

$$V[X] = \sigma^2$$

Why Gaussian distribution is popular



Properties

1) Mathematical simplicity → can be used for easy computation

2) Central Limit Theorem

Central Limit Theorem

multiple R.Vs following diff P.D. can be combined to give Gaussian P.D.

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Random vector

assume n R.Vs.

$$X_1, X_2, \dots, X_n$$

$$X = [X_1 \dots X_n]$$

↑

~~Random vector~~

Random vector

univariate G.D. $\rightarrow$ single variable/vector.

Multivariate Gaussian Distribution